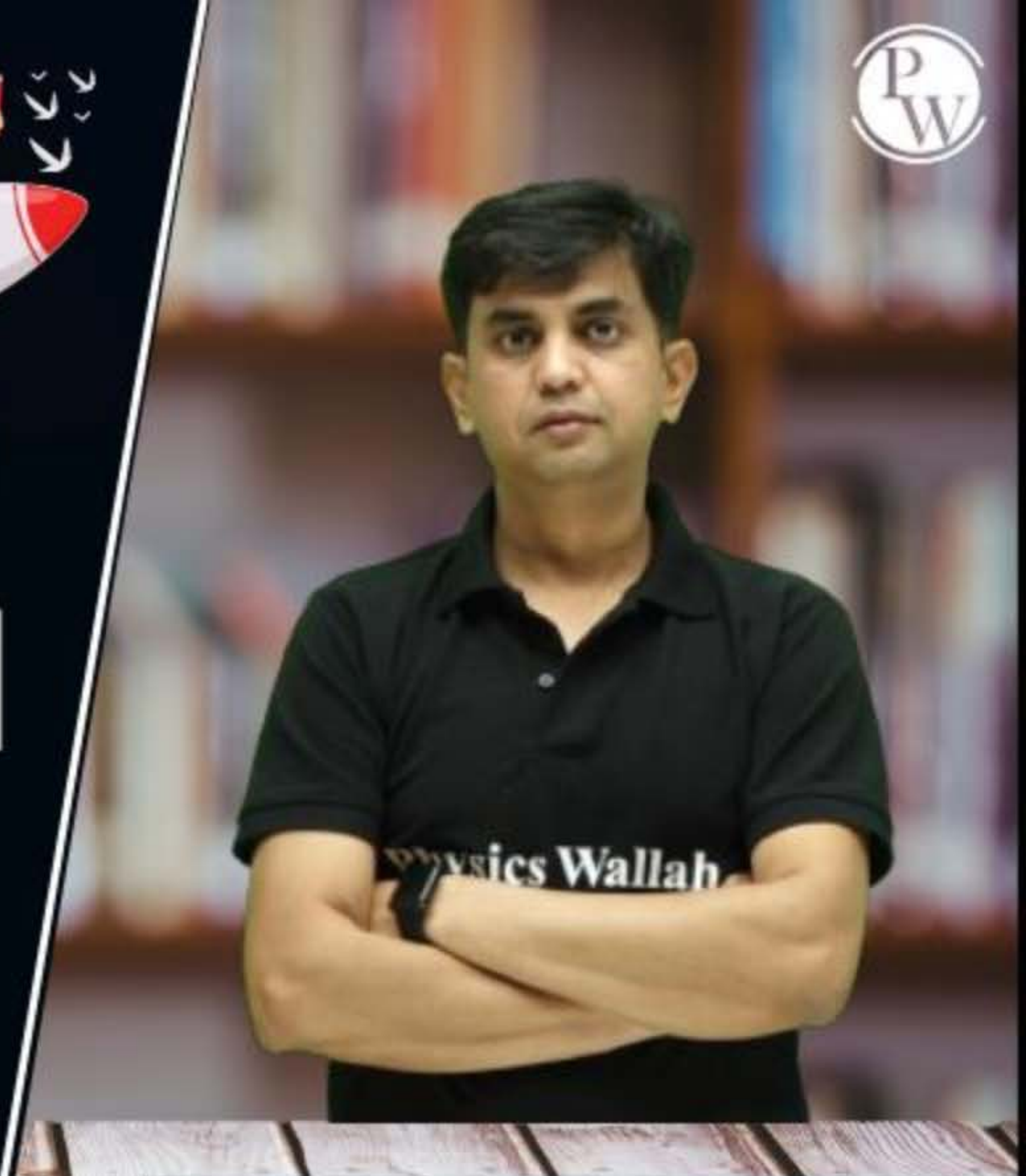
CS & IT ENGINEERING

Computer Network

Introduction

Lecture No. - 07



By - Abhishek Sir

Recap of Previous Lecture

Topic Network Topology

Topic Circuit Switching

Topics to be Covered

Topic Packet Switching

Topic Virtual Circuit Switching

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Message Switching

→ Application processes doing communication by exchanging "messages"

Advantage :-

→ No any dedicated path required between sender and receiver (Unlike circuit switching)

→ Store and Forward

Disadvantage :-

→ Entire message is transmitted as single unit

Topic : Packet Switching

→ Message is divided into smaller size packets ⇒ optimum packet size

[Packets may be same or different size]

→ Example : Internet is a packet-switched network

→ is a datagram network

Topic : Packet Switching → TCP/IP Model

Advantage :- ✓

→ Store and Forward [Datagram Network] → OSI Model

[No any established circuit required between sender and receiver]

→ Efficent utilization of network resources

[Lead to better utilization of bandwidth resource]

(V. Imp.)

(in compare to circuit switching)

→ Inexpensive

Topic : Packet Switching

Disadvantage :-

→ Congestion may occur $\boxed{\text{during routing}}$

→ Every packet is treated $\boxed{\text{independently at every intermediate router}}$

=> More per packet processing overhead at intermediate router

[in compare to circuit switching]

=> Packets may follow different routing paths

=> Packets may have different end-to-end delay

Topic : Types of services

→ Based on order of delivery of data (or packets) at reciver

→ Types of network services :

1. Connection Oriented Services
   [Order of delivery of data (or packets) is same as transmitter transmitted]

2. Connection Less Services
   [Data (or packets) can be delivered in any order to receiver]

Topic : Types of services

→ Circuit switching provide Connection Oriented and Reliable services

→ Packet switching provide Connection Less and Unreliable services

→ Sometimes packet switching may require reordering of packets at receiver

Reliable: No any data (or packet) lost

Unreliable: Packets may lost in the n/w (due to cong)

IP : Connection less
Unreliable
Best Effort Delivery.

IP → Packet-switch protocol

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* Packet switching is faster than circuit switching.

$\boxed{\text{City}_1}$ $\boxed{\text{City}_2}$

#Q. Which one of the following statements is FALSE?

IIT-D

[GATE 2004, 1-Mark]

(A) Packet switching leads to better utilization of bandwidth resources than circuit switching TRUE

(B) Packet switching results in less variation in delay than circuit switching (FALSE)

(C) Packet switching requires more per-packet processing than circuit switching (TRUE)

(D) Packet switching can lead to reordering unlike in circuit switching TRUE

$\boxed{\text{Ans: B}}$

Topic : Virtual Circuit Switching

→ Need to establish virtual circuit (VC) between sender and receiver before transmission over packet-switch network

→ 3 phase in virtual circuit switching :

1. VC setup
2. Data Transfer
3. VC teardown

VC → SVC (switched)
VC → PVC (permanent)

→ Entire routing path of packets is determined before transmission [Entire routing path is fixed for duration of virtual circuit]

→ Example : Used in technologies like X.25, Frame Relay, and ATM

Topic : Virtual Circuit Switching

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Advantage :-

→ Every packets follow each other on predefined path
[Connection Oriented Packet Switching]

Disadvantage :-

→ Congestion may occur during routing

→ Packets may have different end-to-end delay

Topic : Virtual Circuit Switching

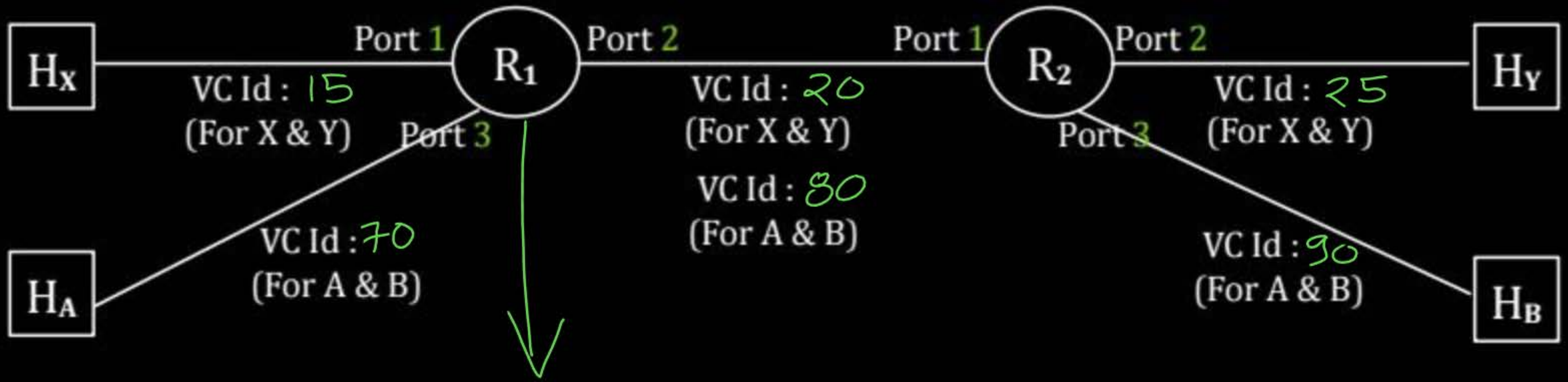
→ Virtual Circuit consists :

1. Path : Series of links and routers, between sender and receiver
2. $\boxed{\text{VC Id}}$: One number for each link along the path
3. Entries in forwarding table in each router along the path

→ Each packet carry VC Id (virtual circuit number) in packet header
[Moving along the path from sender to receiver, no any IP Address]

→ VC Id in packet header, updated by each intermediate router along the path

Topic : Virtual Circuit Switching



Router 1, Forwarding table for Virtual circuit switching

Input Port No.	Input VC Id	Output Port No.	Output VC Id
1	15	2	20
2	20	1	15
3	70	2	80
2	80	3	70

Topic : Virtual Circuit Switching

→ 3 phase in virtual circuit switching :

1. VC setup : Call request and call accept packets
   - → Transport layer specify receiver's IP Address
   - → Network layer determines the path between sender and receiver
   - → Also determine VC Id for each link along the path
   - → Adds an entry in the forwarding table in each router
2. Data Transfer : One number for each link along the path
3. VC teardown :
   - → Sender (or receiver) initiate call termination
   - → Update the forwarding table in each router along the path

2 mins Summary

Topic Packet Switching

Topic Virtual Circuit Switching

THANK - YOU

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